

## Technology Stack

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|----------------------|---------------|
| <b>Date</b>          | 15 July 2026  |
| <b>Team ID</b>       |               |
| <b>Project Name</b>  | Book A Doctor |
| <b>Maximum Marks</b> | 3 Marks       |

## Define Your Technology Stack

A well-chosen technology stack ensures that the Book A Doctor application is scalable, maintainable, and aligned with healthcare project requirements. The following table outlines the technologies selected for each layer of the architecture, along with justifications.

### Technology Stack Details

| S.No | Architecture Component / Layer | Technology Chosen                                         | Justification / Purpose                                                                                 |
|------|--------------------------------|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| 1    | Frontend / Client-Side         | React.js + HTML5/CSS3 + Bootstrap                         | Provides responsive design, reusable components, and a modern UI for patients and doctors.              |
| 2    | Backend / Server-Side          | Node.js with Express.js                                   | Fast, scalable, and lightweight; supports REST APIs; widely used for healthcare web apps.               |
| 3    | Database / Data Storage        | MongoDB                                                   | Handles patient records and appointment data efficiently; supports flexible schema for healthcare data. |
| 4    | Cloud / Hosting / Deployment   | AWS (EC2, S3) + Docker                                    | Ensures easy scalability, secure hosting, and containerized deployment for reliability.                 |
| 5    | Version Control & CI/CD        | GitHub + GitHub Actions                                   | Enables seamless collaboration, version control, and automated testing/deployment.                      |
| 6    | Third-Party APIs / Other Tools | Twilio (SMS reminders), Google Maps API (clinic location) | Twilio sends appointment reminders; Google Maps helps patients locate nearby doctors.                   |